Node.js Knowledge Base

Origin of Node.js

Node.js was created by Ryan Dahl and first released in 2009. It lets developers run JavaScript outside the browser, on servers and in command-line tools.

The V8 Engine

Node.js is built on the V8 JavaScript engine, the same engine that powers Google Chrome. V8 compiles JavaScript directly into machine code for speed.

The Event Loop

Node.js runs JavaScript on a single thread and uses an event loop to handle many connections at once. Slow operations such as reading files or making network calls are non-blocking, so the thread is never stuck waiting.

libuv and the Thread Pool

The libuv library provides the event loop and a thread pool. By default the libuv thread pool has 4 threads, which are used for tasks like file system operations and DNS lookups.

npm

npm is the default package manager for Node.js. It downloads packages from the npm registry using commands like npm install.

package.json and the Lock File

The package.json file describes a project, its scripts and its dependencies. The package-lock.json file records the exact version of every installed dependency so installs are repeatable.

Module Systems

Node.js supports two module systems. CommonJS uses require and module.exports, while ES modules use import and export. To use ES modules, set "type": "module" in package.json.

Working with Files

The fs module reads and writes files. The promise-based API is available as fs.promises, which works well with async and await.

Building Web Servers

The built-in http module can create a web server, but most developers use Express, a minimal web framework. Express adds routing and middleware on top of http.

Express Middleware

In Express, middleware is a function that receives req, res and next. Calling next() passes control to the next middleware in the chain.

process.nextTick

The function process.nextTick schedules a callback to run right after the current operation finishes, before the event loop continues and before promise callbacks.

setImmediate

The setImmediate function runs a callback in the check phase of the event loop, after I/O events. Using setTimeout with a delay of 0 runs the callback in the timers phase instead.

Streams

Streams process data piece by piece instead of loading everything into memory. There are four types: Readable, Writable, Duplex and Transform.

Buffers

A Buffer is a fixed-size chunk of memory used to handle raw binary data such as images or TCP streams.

Events

The events module provides the EventEmitter class. Objects register listeners with on and trigger them with emit.

Using All CPU Cores

The cluster module forks multiple worker processes that share the same server port, so an application can use all CPU cores.

CPU-Heavy Work

For CPU-heavy work such as image processing, the worker_threads module runs JavaScript in separate threads without blocking the main event loop.

Release Lines

Node.js releases even-numbered major versions as Long Term Support (LTS) versions. LTS versions get bug fixes and security updates for about 30 months, so they are recommended for production.

Configuration

Configuration such as API keys is read from process.env. The dotenv package loads variables from a .env file into process.env.

Async Code

The async/await syntax is built on top of Promises and lets asynchronous code read like synchronous code. An async function always returns a Promise.

Paths

The path module builds file paths that work on every operating system. For example, path.join combines segments using the correct separator.

Error Handling

In async code, wrap await calls in try/catch to handle errors. An unhandled promise rejection can crash a Node.js process in modern versions.